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Professional Summary

Full-Stack Engineer focused on building reliable, production-grade web platforms. Experienced with Django REST Framework, React, PostgreSQL, Docker, and AWS to design secure APIs, scalable backend systems, and high-performance applications. Emphasizes system reliability, automation, and real-world operational constraints through CI/CD pipelines, payment integrations, and cloud deployment, delivering software that operates correctly in production environments.

Experience

- **Full Stack Developer — Bridgeon** May 2025 – Present
  - Architected and delivered scalable web applications using Django REST Framework, React, and PostgreSQL following modular service-oriented design principles.
  - Designed secure REST APIs implementing JWT authentication, role-based access control (RBAC), and layered permission enforcement.
  - Improved backend performance by optimizing database queries and request execution paths, reducing API response latency by up to 35%.
  - Built state-driven admin interfaces using React + Redux Toolkit, ensuring predictable data flow and reducing UI inconsistency bugs.
  - Implemented Razorpay payment workflows with transaction verification, idempotent order handling, and failure recovery mechanisms.
  - Deployed applications on AWS (EC2, RDS) with Nginx + Gunicorn, implementing reverse proxying, process management, and zero-downtime deployment strategies via CI/CD pipelines.
  - Live demo: https://noirel-perfume.vercel.app/
- **Project Intern — Regional Technologies** Oct 2024 – Mar 2025
  - Built a cross-platform digital governance platform using Django backend services and a Flutter mobile application to streamline citizen–administration interaction.
  - Modeled hierarchical governance architecture (District → Higher Authority → Panchayath → Citizens) with role-based permission controls.
  - Developed REST APIs supporting citizen services such as service requests, complaint submission, notifications, and community updates.
  - Implemented complaint escalation workflows allowing unresolved Panchayath-level issues to be forwarded to higher administrative authorities.
  - Built an announcement and update feed enabling Panchayath administrators to publish news, alerts, and public service information to residents.
  - Enabled citizen engagement features including community posts, media sharing, and interaction within local governance networks.
  - Designed normalized and query-efficient data models, reducing redundant joins and improving query performance under relational workloads.

Technical Skills

- **Languages:** Python, JavaScript, Dart
- **Backend Development:** Django, Django REST Framework, REST API Design, Authentication (JWT), Role-Based Access Control (RBAC), API Security
- **Frontend Development:** React.js, Redux Toolkit, Component-Based Architecture, API Integration
- **Databases:** PostgreSQL, MySQL, Query Optimization
- **Cloud & DevOps:** AWS (EC2, RDS, S3), Docker, Nginx, Gunicorn, GitHub Actions CI/CD
- **Engineering Practices:** Performance Optimization, Scalable System Design, Clean Architecture

Engineering Highlights

- Implemented debouncing and throttling techniques to optimize client-side event handling and reduce redundant API calls.
- Designed resilient UI architectures using React Error Boundaries with fallback components.
- Applied performance-focused frontend patterns including lazy loading, shimmer slides, memoization strategies, and efficient state updates.

Projects

- **Cipher Analytics — Privacy-Preserving Distributed AI Platform (Ongoing)**
  - Architecting a privacy-first analytics system enabling secure computation on encrypted datasets using Fully Homomorphic Encryption (CKKS via TenSEAL/SEAL), supporting encrypted ML inference without data exposure.
  - Designed a multi-queue distributed processing architecture with Celery + Redis, isolating FHE, ML, and I/O workloads to optimize throughput, latency, and fault tolerance under heavy compute loads.
  - Built an explainable ML pipeline using SHAP for feature attribution, and applied UMAP + HDBSCAN to transform high-dimensional data into interpretable clusters and anomaly patterns.
  - Built real-time, interactive data visualization layer using WebGL (Three.js + React Three Fiber) and ECharts for large-scale statistical and 3D exploratory analytics.
  - Engineered asynchronous, event-driven backend with Django (DRF), Channels (WebSockets), and Daphne for live task tracking and system feedback loops.
  - Enforced production-grade security and observability including rate limiting, audit trails (django-simple-history), static typing (mypy), and security linting (bandit).
  - Containerized and orchestrated services using Docker with shared-volume architecture for efficient inter-service data access in distributed environments.
  - Optimized distributed task execution by isolating compute-heavy workloads, reducing processing latency under concurrent load scenarios.
- **NOIRÉL — Luxury E-Commerce Platform (Production)**
  - Built a full-stack e-commerce platform using React and Django REST Framework supporting the complete order lifecycle from product browsing to payment confirmation.
  - Designed modular REST APIs for catalog management, cart operations, order processing, and payment verification.
  - Implemented JWT-based authentication with protected routes, admin control panel, and role-based access control.
  - Integrated Razorpay payment gateway with secure transaction validation, webhook verification, and failure recovery handling.
  - Optimized frontend performance with efficient state management and lazy-loaded components, achieving a **100/100 Lighthouse performance score**.
  - Deployed production environment on AWS infrastructure using Nginx + Gunicorn with automated CI/CD pipelines through GitHub Actions.
- **Blog System Design Sandbox (AWS Deployed)**
  - Engineered a backend architecture sandbox to experiment with production-grade API design, authentication flows, and failure handling patterns.
  - Implemented secure authentication with access/refresh JWT separation, token versioning, and forced token invalidation on password reset.
  - Designed consistent API response envelopes, error handling middleware, and request validation patterns.
  - Dockerized frontend and backend services to ensure environment parity across development, staging, and production.
  - Implemented structured logging, request-scoped trace IDs, soft-delete patterns, and application health-check endpoints.
  - Automated CI/CD workflows with GitHub Actions enabling build validation and deployment consistency.
- **Civic (CiviTech) — Cross-Platform Digital Governance Platform**
  - Developed a multi-role civic engagement platform connecting citizens with Panchayath and higher administrative authorities.
  - Architected hierarchical governance data models representing District → Higher Authority → Panchayath → Citizen relationships.
  - Implemented citizen service modules including service requests, verified announcements, emergency alerts, and public updates.
  - Designed complaint escalation workflows enabling unresolved Panchayath-level issues to be forwarded to higher authorities.
  - Built administrative dashboards supporting moderation, analytics, and service monitoring for local governance bodies.
  - Enabled citizen interaction through community feeds, media sharing, and discussion within local governance networks.

Education

- **Bachelor of Computer Applications (BCA)** — SAFA Arts & Science College Aug 2022 – Apr 2025

Certifications

- Python Django & Flutter Certification — Regional Technologies (2025)
- Certified Cyber Security Analyst (CCSA) — Red Team Hacker Academy (2022)